HR Attrition Analysis

Narrative Report & Actionable Recommendations

1. Executive Summary

This report analyzes workforce attrition patterns across 1,470 employees using the IBM HR Analytics dataset. The overall attrition rate stands at 16.1%, meaning 237 employees out of 1,470 have left the organization. While this figure alone may seem manageable, deeper analysis reveals concentrated risk pockets — specific job roles, departments, and behavioral patterns — where attrition rates are significantly above average, posing real operational and financial risks.

Overall Attrition Rate: 16.1% — 237 out of 1,470 employees left the organization

The analysis identifies three primary attrition drivers: excessive overtime demands, compensation gaps between leavers and active employees, and early-tenure disengagement. Each of these is actionable through targeted HR interventions.

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2. Key Findings

2.1 Overall Attrition Snapshot

Metric	Value	Benchmark Context
Total Employees	1,470	Full dataset analyzed
Employees Who Left	237	16.1% of total workforce
Avg. Income — Active	$6,833 / mo	Baseline for comparison
Avg. Income — Leavers	$4,787 / mo	29.9% lower than active
Avg. Tenure — Leavers	5.1 years	vs. 7.4 years for stayers

2.2 Finding 1 — Overtime is the Strongest Attrition Predictor

Employees who work overtime show an attrition rate of 30.5%, nearly three times higher than employees without overtime (10.4%). This is the single strongest predictor of attrition in the entire dataset — stronger even than salary level or job satisfaction scores.

- Overtime = Yes: 30.5% attrition rate
- Overtime = No: 10.4% attrition rate
- Risk multiplier: 2.9x higher risk for overtime employees

This suggests that workload imbalance and burnout are significant contributors to voluntary turnover, and addressing overtime distribution could yield the highest return on retention investment.

2.3 Finding 2 — Sales Department and Sales Representative Role are Clear Outliers

At the department level, Sales shows the highest attrition at 20.6%, followed by Human Resources at 19.0%, while Research & Development is meaningfully lower at 13.8%. Within Sales, the Sales Representative job role stands out as the single biggest outlier in the entire dataset.

- Sales Representative: 39.8% attrition — 2.5x the company average
- Laboratory Technician: 23.9%
- Human Resources: 23.1%
- Research Director: 2.5% — the lowest in the company (positive outlier)

The Sales Representative role losing nearly 4 in every 10 employees is a critical operational risk. Given that sales roles typically require long ramp-up periods, the constant turnover in this position directly impacts revenue generation capacity.

2.4 Finding 3 — Compensation Gap Signals Pay Equity Issue

Employees who left the organization earned on average $4,787 per month, compared to $6,833 for those who stayed — a gap of approximately 30%. While correlation does not imply causation, this consistent pattern across the dataset suggests that compensation competitiveness plays a meaningful role in retention decisions.

- Active employees’ average monthly income: $6,833
- Leavers average monthly income: $4,787
- Income gap: $2,046 (29.9% lower for leavers)

2.5 Finding 4 — Early Tenure is a Critical Vulnerability Window

Employees in their first year show an attrition rate of 34.9%, which drops steadily as tenure increases. By the time employees reach 11+ years, attrition falls to just 8.1%. This pattern clearly shows that the highest-risk period is the first 1-3 years of employment — the onboarding and early engagement phase.

- Tenure 0-1 year: 34.9% attrition
- Tenure 2-3 years: 18.4%
- Tenure 4-5 years: 13.1%
- Tenure 6-10 years: 12.3%
- Tenure 11+ years: 8.1%

2.6 Finding 5 — Young Employees and Frequent Travelers Are High-Risk Groups

Employees aged 18-25 show the highest attrition rate at 35.8%, significantly above older age groups. Similarly, employees who travel frequently for business show 24.9% attrition versus just 8.0% for non-travelers. These two groups likely overlap substantially with the Sales Representative role, compounding the risk concentration.

- Age 18-25: 35.8% attrition
- Age 26-35: 19.1%
- Travel Frequently: 24.9% vs. Non-Travel: 8.0%
- Single employees: 25.5% vs. Married: 12.5%

3. Actionable Recommendations

Based on the five key findings above, the following four recommendations are prioritized by potential impact and feasibility. Each recommendation targets a specific, measurable driver identified in the data.

#	Recommendation	Target Group	Expected Impact
1	Introduce Overtime Monitoring & Rotation Policy: Set a monthly overtime cap per employee and implement workload-sharing rotations to prevent burnout concentration. Flag employees exceeding the cap for immediate manager check-in.	All employees, especially Sales & R&D	Reduce overtime-driven attrition from 30.5% toward the baseline 10.4%. Potential to retain ~50 employees annually.
2	Conduct Immediate Compensation Review for Sales Representatives: Benchmark Sales Rep salaries against market data. Introduce performance-linked compensation bands that close the ~$2,000/month gap between leavers and stayers in this role.	Sales Representative role (39.8% attrition)	Most direct lever for the highest-risk role. A 10% salary increase for this group could yield significant ROI given hiring and training costs.
3	Implement a Structured 90-Day & 1-Year Onboarding Program: Assign dedicated mentors to all new hires for their first year. Conduct mandatory check-in surveys at 30, 60, 90, and 180 days to identify disengagement signals early.	New hires (0-2 years tenure)	Address the 34.9% first-year attrition rate. Early intervention is significantly cheaper than replacement (avg. cost to replace = 50-200% of annual salary).
4	Create Career Development Tracks for Young Employees: Develop transparent promotion criteria and fast-track programs for the 18-25 age group. Pair with mentorship from Senior/Director-level employees who show the lowest attrition rates.	Employees aged 18-25 (35.8% attrition)	Converts highest-risk demographic into long-term contributors. Addresses the lack of perceived growth that often drives early-career exits.

4. Business Impact Estimate

To contextualize the financial significance of attrition, consider the following conservative estimate:

- Industry standard replacement cost: 50%–150% of annual salary per employee
- Average monthly income of leavers: $4,787 → Annual: ~$57,444
- Conservative replacement cost per leaver: ~$28,722 (50% of annual salary)
- Total estimated annual cost of current attrition (237 employees): ~$6.8 million

Even reducing attrition by 5 percentage points (from 16.1% to 11.1%) would retain approximately 73 additional employees annually, representing an estimated cost saving of over $2 million — a conservative return on any targeted retention investment.

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5. Conclusion

The data tells a clear story: attrition at this organization is not uniformly distributed. It is concentrated in specific roles (Sales Representative), specific behaviors (overtime workers), and specific life stages (new hires, young employees).

This concentration is actually good news — it means targeted interventions can produce outsized results without organization-wide restructuring.

The four recommendations in this report are designed to be implemented incrementally, with measurable outcomes trackable through the same Power BI dashboard built for this analysis. Attrition Rate by Department and Job Role should be monitored monthly as the primary KPI for all retention initiatives.